

FPGA ASSESSMENT 6

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1. Aim

Design a **256 × 6-bit dual-port RAM** using the Quartus Prime IP Catalog. The memory must support two ports, A and B, with an 8-bit address for each port and a 6-bit data path. The design is then implemented on the DE2-115 FPGA board. Switches provide the address and data inputs, push buttons provide controlled data capture/read actions, and the on-board seven-segment displays show the two port addresses and the corresponding data values.

2. Quartus Prime RAM IP Generation

The RAM is first generated through the Quartus Prime IP Catalog; the generated IP is then instantiated from the user-written top-level module. The supplied screenshots document the sequence.

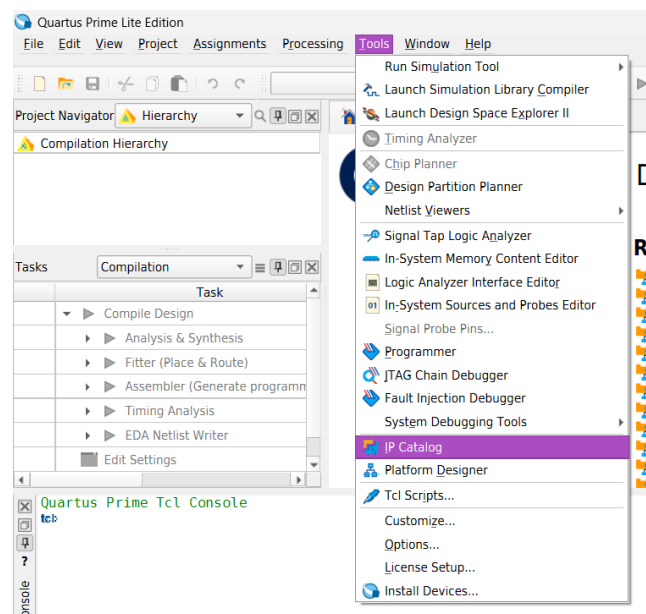


Figure 1: Opening the Quartus Prime Tools menu and selecting the IP Catalog.

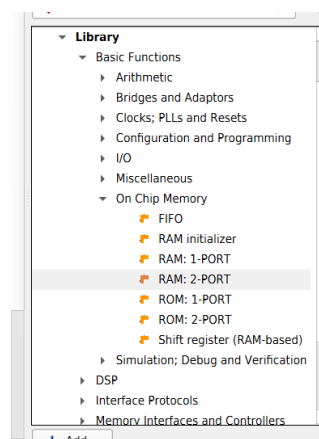


Figure 2: Selecting RAM: 2-PORT under Basic Functions → On Chip Memory in the IP Catalog.

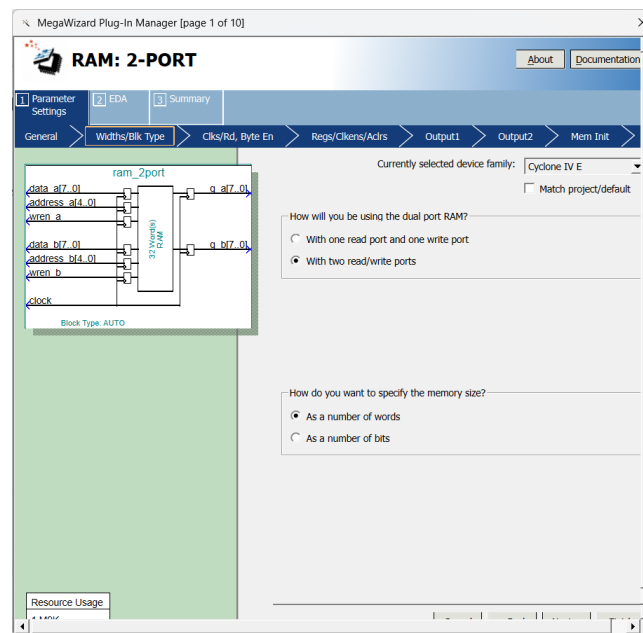


Figure 3: MegaWizard parameter page showing the Cyclone IV E family and two read/write ports.

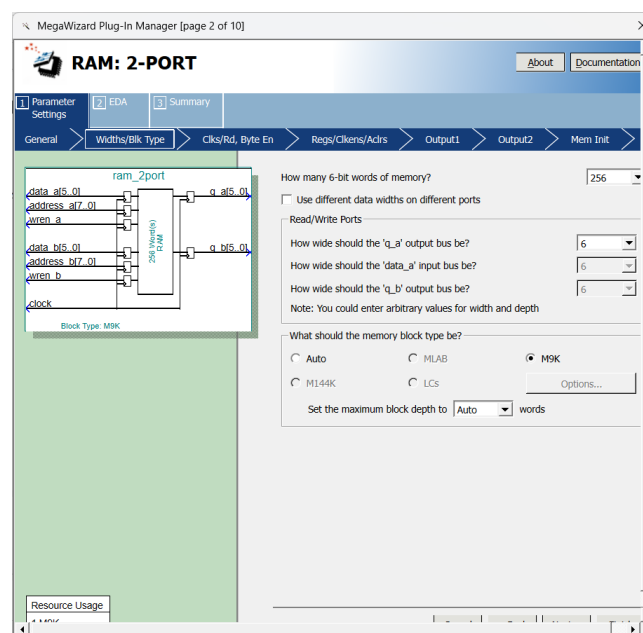


Figure 4: RAM width/depth configuration: 256 words, 6-bit data width and M9K block type.

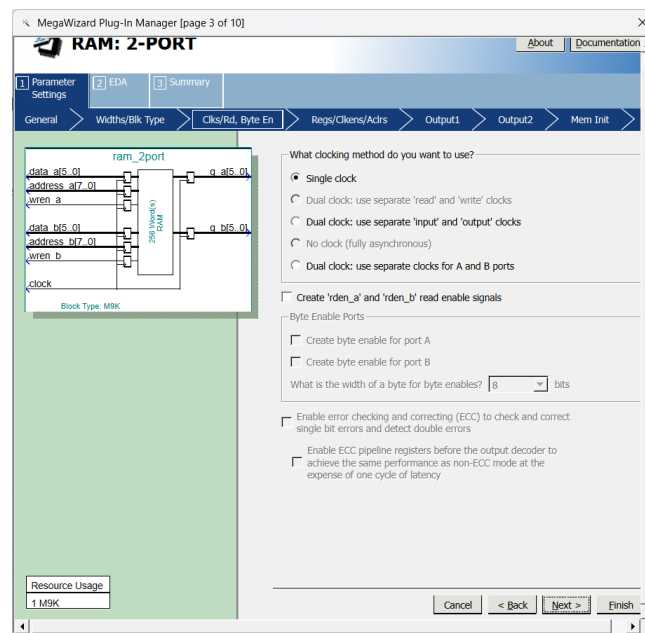


Figure 5: Clock configuration showing a single shared clock.

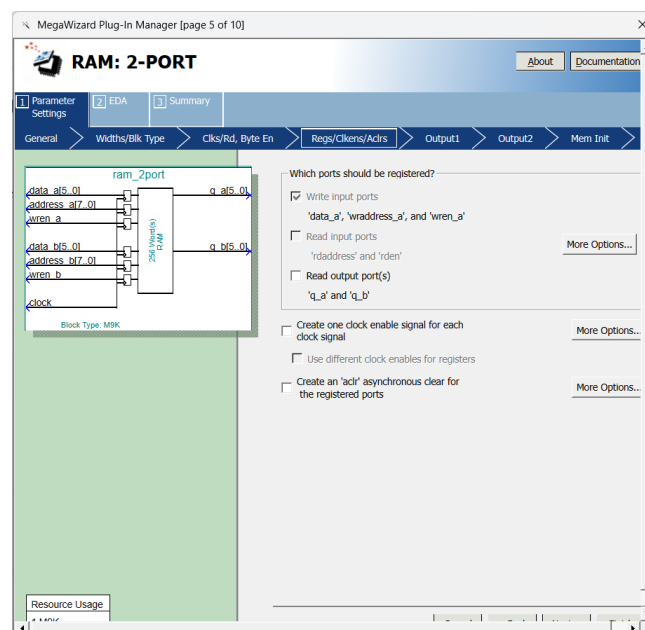


Figure 6: Register/clock configuration with write input ports selected.

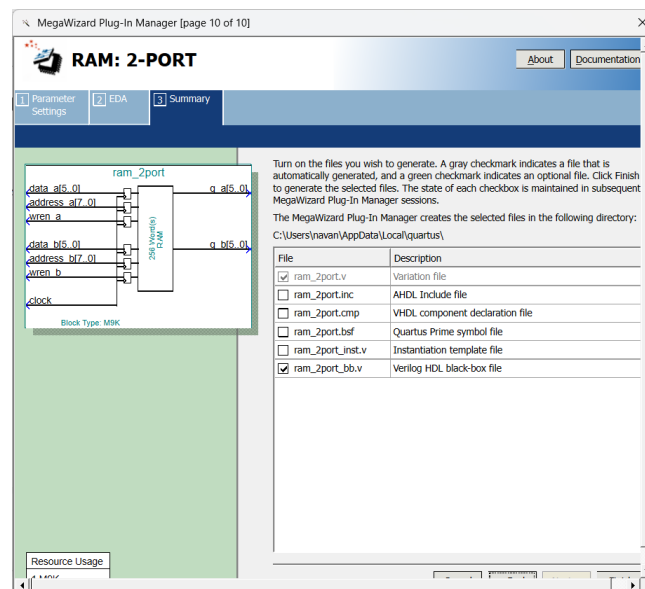


Figure 7: MegaWizard summary page showing generation of `ram_2port.v` and `ram_2port_bb.v`.

3. Experimental Procedure

1. Create/open the Quartus Prime project for the DE2-115 Cyclone IV E device.
2. Open **Tools** → **IP Catalog** and select **RAM: 2-PORT**.
3. Configure the IP for two read/write ports, 256 words and 6-bit data width, using an M9K block and a single clock.
4. Generate the IP files `ram_2port.v` and `ram_2port_bb.v`.
5. Add the generated RAM files and the user HDL modules to the project.
6. Set `ram_2port_top` as the top-level entity.
7. Apply the supplied `pin_assignment.tcl` constraints and compile the project.
8. Program the DE2-115 FPGA through the Quartus Programmer.
9. In write mode, set the A/B addresses using the switch groups, enter each 6-bit data value using `SW[5:0]`, and capture the A/B data with KEY0/KEY1. Press KEY2 to commit the memory write.
10. Switch to read mode, select the stored A/B addresses, and press KEY0/KEY1 to capture and display the values read from ports A and B.

4. FPGA Hardware Implementation and Results

The submitted photographs show the actual DE2-115 board after programming. They document the address setup, individual data-writing steps, the mode change and the subsequent readback of port A and port B values.

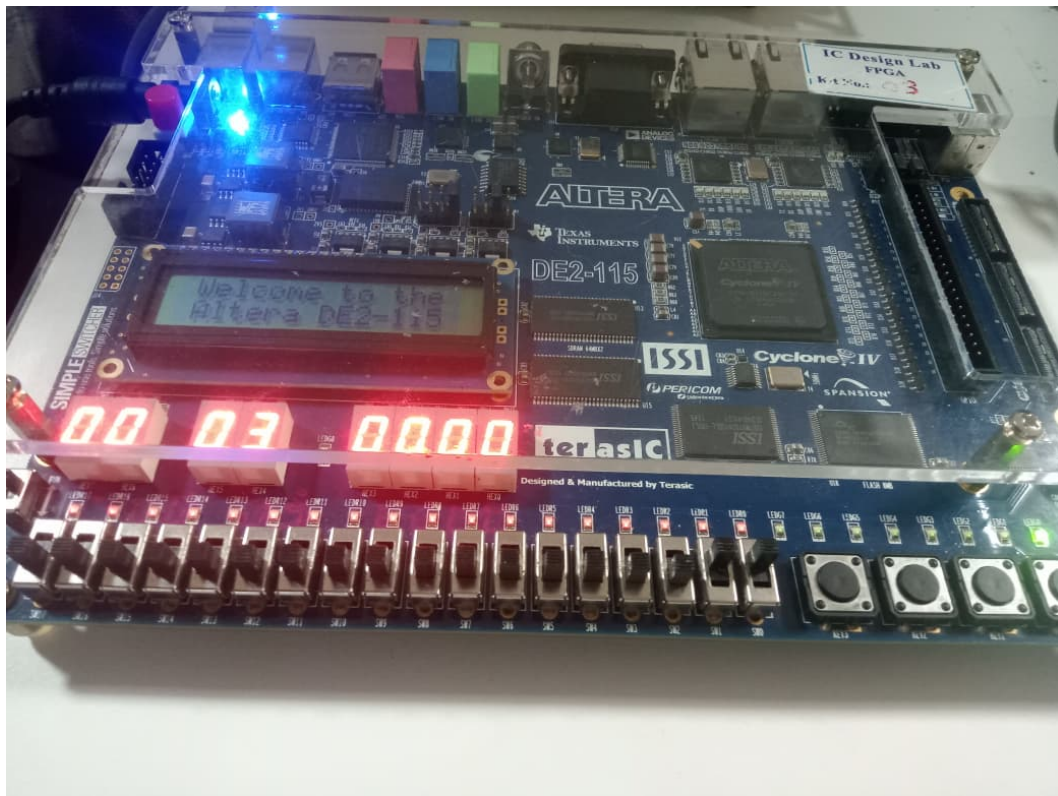


Figure 8: FPGA implementation during the Port-A data-writing step. The address/data display and active controls are visible on the DE2-115 board.

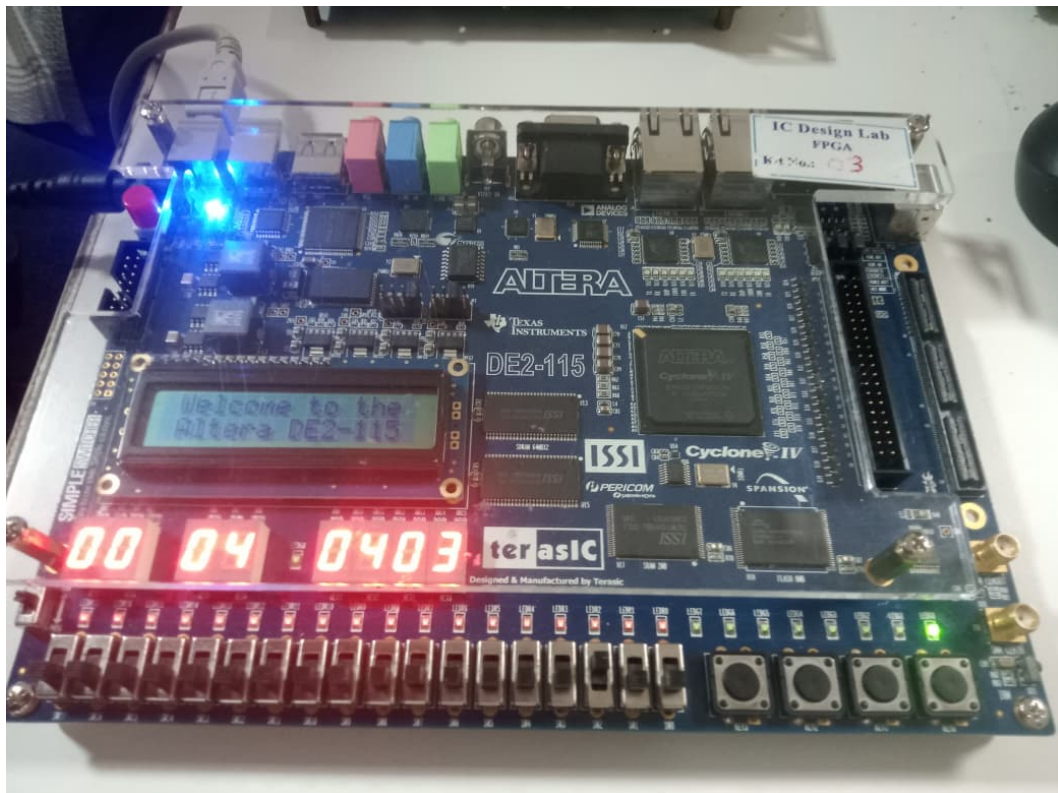


Figure 9: FPGA implementation during the Port-B data-writing step.

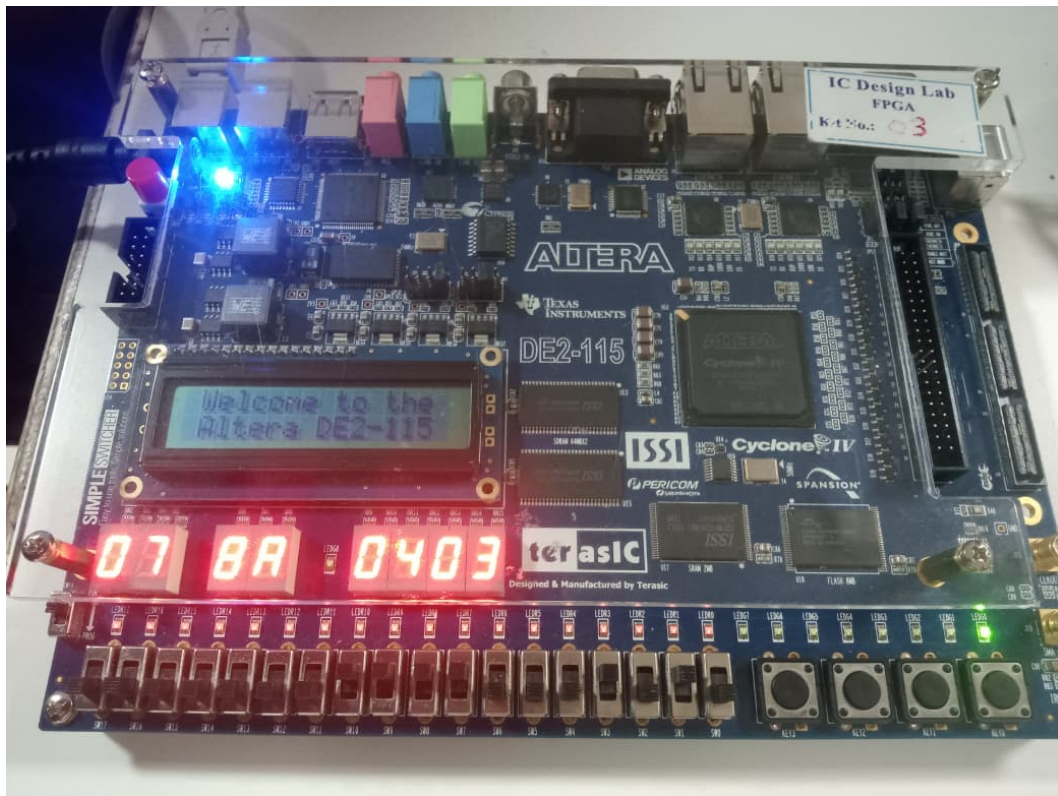


Figure 10: FPGA implementation with the A and B address fields set for the RAM operation.

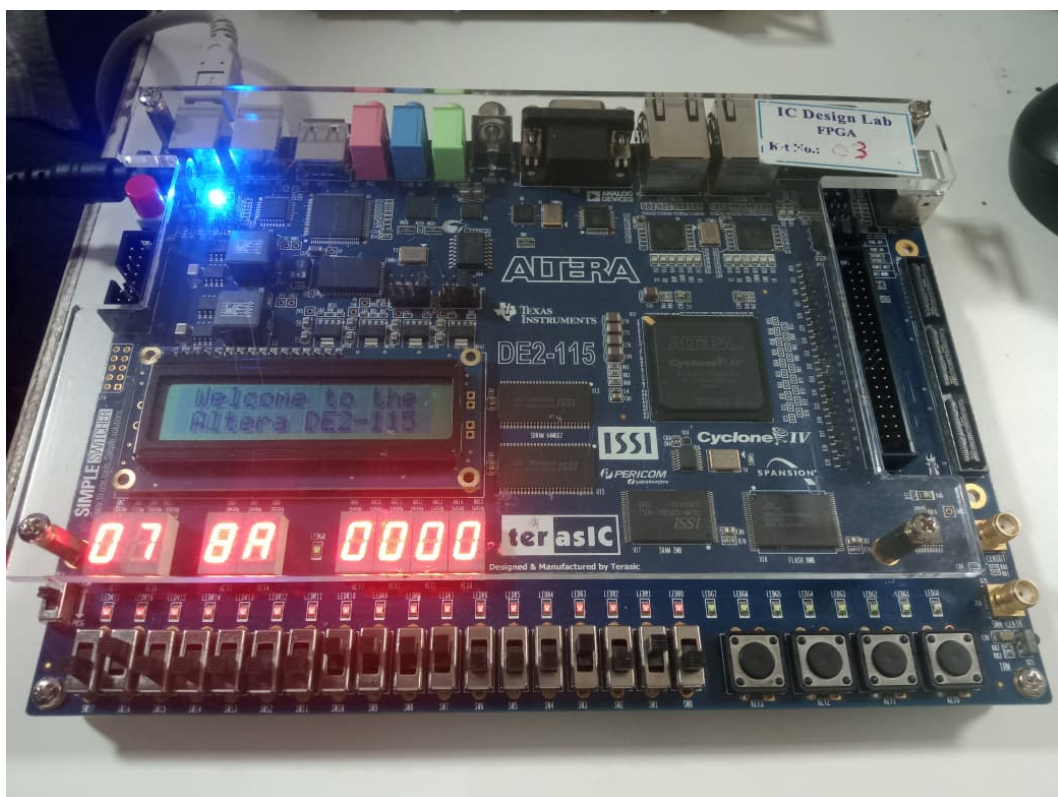


Figure 11: FPGA implementation after switching to read mode.

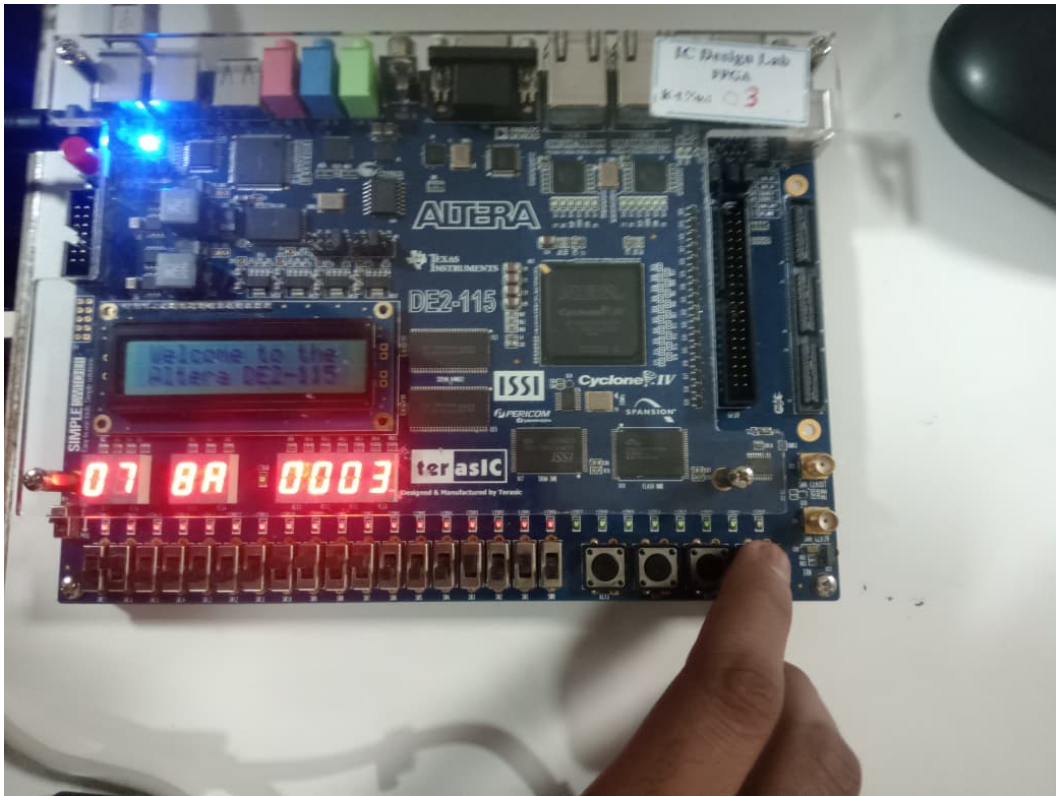


Figure 12: FPGA readback of Port A using the push-button controlled capture.

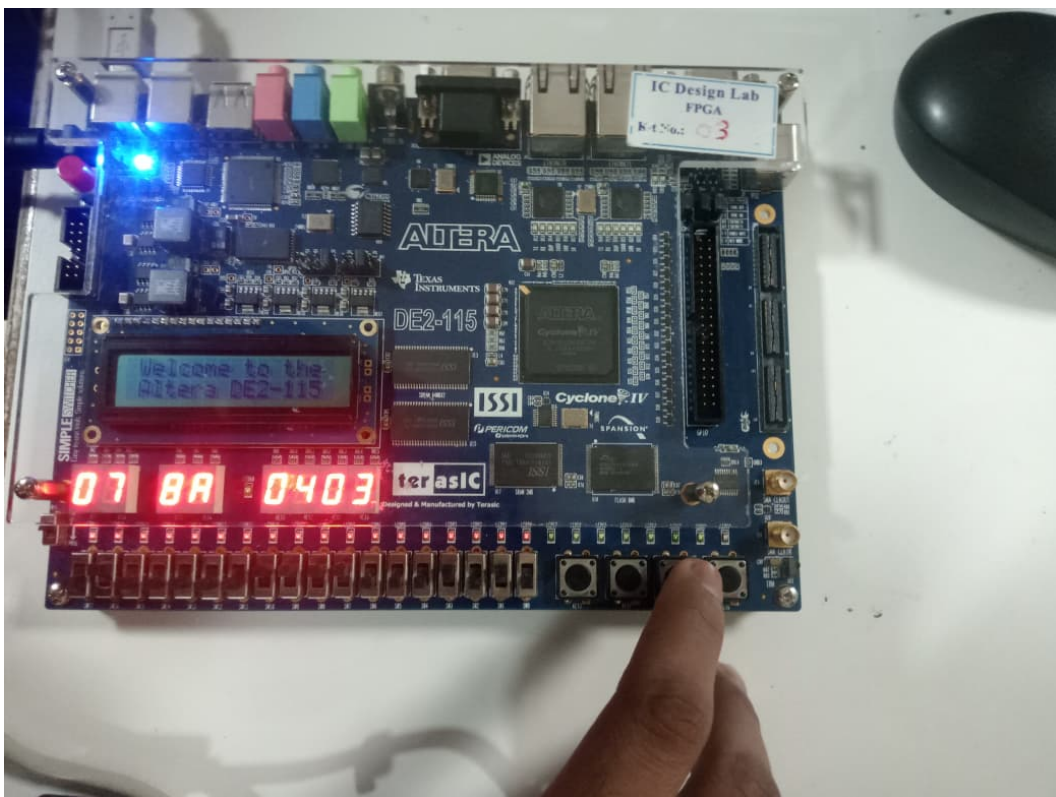


Figure 13: FPGA readback of Port B using the push-button controlled capture.

A. Verilog Code

A.1. Top-Level Module: ram_2port_top.v

```

1 module ram_2port_top(CLOCK_50,SW,KEY,HEX0,HEX1,HEX2,HEX3,HEX4,HEX5,HEX6,HEX7,LEDG0);
2
3 input  CLOCK_50;
4 input  [17:0]SW;
5 input  [3:0]KEY;
6
7 output [6:0]HEX0,HEX1,HEX2,HEX3,HEX4,HEX5,HEX6,HEX7;
8 output LEDG0;
9
10 wire rst;
11 wire write_mode;
12 wire read_mode;
13
14 wire k0_pulse,k1_pulse,k2_pulse,k3_pulse;
15
16 wire [7:0]address_a;
17 wire [7:0]address_b;
18
19 wire [5:0]data_a;
20 wire [5:0]data_b;
21
22 wire [5:0]q_a;
23 wire [5:0]q_b;
24
25 reg [5:0]data_a_reg;
26 reg [5:0]data_b_reg;
27
28 reg [5:0]read_a_reg;
29 reg [5:0]read_b_reg;
30
31 wire [5:0]display_a;
32 wire [5:0]display_b;
33
34 wire wren_a;
35 wire wren_b;
36
37 assign rst = SW[17];
38 assign write_mode = ~SW[16];
39 assign read_mode = SW[16];
40
41 // Separate address switch groups
42 // Port A address = SW[7:0]
43 // Port B address = SW[15:8]
44 assign address_a = SW[7:0];
45 assign address_b = SW[15:8];
46
47 assign data_a = data_a_reg;
48 assign data_b = data_b_reg;
49
50 // A single physical 6-bit data entry field is used sequentially.
51 // K0 captures data for Port A; K1 captures data for Port B.
52 // This is necessary because DE2-115 has only 18 user switches,
53 // while two independent 8-bit addresses + two independent 6-bit
54 // data buses would require 28 switches.
55
56 assign wren_a = write_mode & k2_pulse;
57 assign wren_b = write_mode & k2_pulse;
58
59 // In read mode, K0/K1 latch the currently addressed RAM outputs.
60 assign display_a = read_mode ? read_a_reg : data_a_reg;
61 assign display_b = read_mode ? read_b_reg : data_b_reg;
62
63 assign LEDG0 = write_mode;
64
65 button_pulse p0(
66     .clk(CLOCK_50),
67     .rst_n(~rst),
68     .key(KEY[0]),
69     .pulse(k0_pulse)
70 );
71

```

```

72 button_pulse p1(
73     .clk(CLOCK_50),
74     .rst_n(~rst),
75     .key(KEY[1]),
76     .pulse(k1_pulse)
77 );
78
79 button_pulse p2(
80     .clk(CLOCK_50),
81     .rst_n(~rst),
82     .key(KEY[2]),
83     .pulse(k2_pulse)
84 );
85
86 button_pulse p3(
87     .clk(CLOCK_50),
88     .rst_n(~rst),
89     .key(KEY[3]),
90     .pulse(k3_pulse)
91 );
92
93 // Data entry and read-back latches
94 always@(posedge CLOCK_50 or posedge rst)
95 begin
96     if(rst)
97     begin
98         data_a_reg<=6'b000000;
99         data_b_reg<=6'b000000;
100        read_a_reg<=6'b000000;
101        read_b_reg<=6'b000000;
102    end
103    else
104    begin
105        if(write_mode)
106        begin
107            if(k0_pulse)
108                data_a_reg<=SW[5:0];
109
110            if(k1_pulse)
111                data_b_reg<=SW[5:0];
112        end
113
114        if(read_mode)
115        begin
116            if(k0_pulse)
117                read_a_reg<=q_a;
118
119            if(k1_pulse)
120                read_b_reg<=q_b;
121        end
122    end
123 end
124
125 // 256 x 6 dual-port RAM
126 // Both ports WRITE together in write mode.
127 // Both ports READ together in read mode.
128 ram_2port r1(
129     .address_a(address_a),
130     .address_b(address_b),
131     .clock(CLOCK_50),
132     .data_a(data_a),
133     .data_b(data_b),
134     .wren_a(wren_a),
135     .wren_b(wren_b),
136     .q_a(q_a),
137     .q_b(q_b)
138 );
139
140 //=====
141 // SEVEN SEGMENT DISPLAY
142 //=====
143 // HEX5 HEX4 = Port A address
144 // HEX1 HEX0 = Port A data/read value
145 // HEX7 HEX6 = Port B address
146 // HEX3 HEX2 = Port B data/read value
147 //=====

```

```

148
149 seven_segment s0(
150     .in(display_a[3:0]),
151     .hex(HEX0)
152 );
153
154 seven_segment s1(
155     .in({2'b00,display_a[5:4]}),
156     .hex(HEX1)
157 );
158
159 seven_segment s2(
160     .in(display_b[3:0]),
161     .hex(HEX2)
162 );
163
164 seven_segment s3(
165     .in({2'b00,display_b[5:4]}),
166     .hex(HEX3)
167 );
168
169 seven_segment s4(
170     .in(address_a[3:0]),
171     .hex(HEX4)
172 );
173
174 seven_segment s5(
175     .in(address_a[7:4]),
176     .hex(HEX5)
177 );
178
179 seven_segment s6(
180     .in(address_b[3:0]),
181     .hex(HEX6)
182 );
183
184 seven_segment s7(
185     .in(address_b[7:4]),
186     .hex(HEX7)
187 );
188
189 endmodule

```

A.2. Button Pulse Module: button_pulse.v

```

1 module button_pulse(clk,rst_n,key,pulse);
2
3 input clk,rst_n,key;
4 output reg pulse;
5
6 reg key1,key2,key2_prev;
7
8 always@(posedge clk or negedge rst_n)
9 begin
10     if(!rst_n)
11     begin
12         key1<=1'b1;
13         key2<=1'b1;
14         key2_prev<=1'b1;
15         pulse<=1'b0;
16     end
17     else
18     begin
19         key1<=key;
20         key2<=key1;
21         key2_prev<=key2;
22         pulse<=1'b0;
23
24         // DE2-115 keys are active-low
25         if(key2_prev && !key2)
26             pulse<=1'b1;
27     end
28 end
29

```

```
30 endmodule
```

A.3. Seven-Segment Decoder: `seven_segment.v`

```
1 module seven_segment(in,hex);
2
3 input [3:0]in;
4 output reg [6:0]hex;
5
6 // DE2-115 HEX displays are common-anode / active-low.
7 // Segment order: {g,f,e,d,c,b,a}
8
9 always@(*)
10 begin
11     case(in)
12         4'd0: hex=7'b1000000;
13         4'd1: hex=7'b1111001;
14         4'd2: hex=7'b0100100;
15         4'd3: hex=7'b0110000;
16         4'd4: hex=7'b0011001;
17         4'd5: hex=7'b0010010;
18         4'd6: hex=7'b0000010;
19         4'd7: hex=7'b1111000;
20         4'd8: hex=7'b0000000;
21         4'd9: hex=7'b0010000;
22         4'd10: hex=7'b0001000; // A
23         4'd11: hex=7'b0000011; // b
24         4'd12: hex=7'b1000110; // C
25         4'd13: hex=7'b0100001; // d
26         4'd14: hex=7'b0000110; // E
27         4'd15: hex=7'b0001110; // F
28         default: hex=7'b1111111;
29     endcase
30 end
31
32 endmodule
```

A.4. Quartus-Generated RAM Core: `ram_2port.v`

```
1 // megafunction wizard: %RAM: 2-PORT%
2 // GENERATION: STANDARD
3 // VERSION: WM1.0
4 // MODULE: altsyncram
5
6 // =====
7 // File Name: ram_2port.v
8 // Megafunction Name(s):
9 //     altsyncram
10 //
11 // Simulation Library Files(s):
12 //     altera_mf
13 // =====
14 // *****
15 // THIS IS A WIZARD-GENERATED FILE. DO NOT EDIT THIS FILE!
16 //
17 // 25.1std.0 Build 1129 10/21/2025 SC Lite Edition
18 // *****
19
20
21 //Copyright (C) 2025 Altera Corporation. All rights reserved.
22 //Your use of Altera Corporation's design tools, logic functions
23 //and other software and tools, and any partner logic
24 //functions, and any output files from any of the foregoing
25 //(including device programming or simulation files), and any
26 //associated documentation or information are expressly subject
27 //to the terms and conditions of the Altera Program License
28 //Subscription Agreement, the Altera Quartus Prime License Agreement,
29 //the Altera IP License Agreement, or other applicable license
30 //agreement, including, without limitation, that your use is for
31 //the sole purpose of programming logic devices manufactured by
32 //Altera and sold by Altera or its authorized distributors. Please
```

```

33 //refer to the Altera Software License Subscription Agreements
34 //on the Quartus Prime software download page.
35
36
37 // synopsys translate_off
38 `timescale 1 ps / 1 ps
39 // synopsys translate_on
40 module ram_2port (
41     address_a,
42     address_b,
43     clock,
44     data_a,
45     data_b,
46     wren_a,
47     wren_b,
48     q_a,
49     q_b);
50
51     input [7:0] address_a;
52     input [7:0] address_b;
53     input clock;
54     input [5:0] data_a;
55     input [5:0] data_b;
56     input wren_a;
57     input wren_b;
58     output [5:0] q_a;
59     output [5:0] q_b;
60 `ifndef ALTERA_RESERVED_QIS
61 // synopsys translate_off
62 `endif
63     tri1 clock;
64     tri0 wren_a;
65     tri0 wren_b;
66 `ifndef ALTERA_RESERVED_QIS
67 // synopsys translate_on
68 `endif
69
70     wire [5:0] sub_wire0;
71     wire [5:0] sub_wire1;
72     wire [5:0] q_a = sub_wire0[5:0];
73     wire [5:0] q_b = sub_wire1[5:0];
74
75     altsyncram altsyncram_component (
76         .address_a (address_a),
77         .address_b (address_b),
78         .clock0 (clock),
79         .data_a (data_a),
80         .data_b (data_b),
81         .wren_a (wren_a),
82         .wren_b (wren_b),
83         .q_a (sub_wire0),
84         .q_b (sub_wire1),
85         .aclr0 (1'b0),
86         .aclr1 (1'b0),
87         .addressstall_a (1'b0),
88         .addressstall_b (1'b0),
89         .byteena_a (1'b1),
90         .byteena_b (1'b1),
91         .clock1 (1'b1),
92         .clocken0 (1'b1),
93         .clocken1 (1'b1),
94         .clocken2 (1'b1),
95         .clocken3 (1'b1),
96         .eccstatus (),
97         .rden_a (1'b1),
98         .rden_b (1'b1));
99     defparam
100         altsyncram_component.address_reg_b = "CLOCK0",
101         altsyncram_component.clock_enable_input_a = "BYPASS",
102         altsyncram_component.clock_enable_input_b = "BYPASS",
103         altsyncram_component.clock_enable_output_a = "BYPASS",
104         altsyncram_component.clock_enable_output_b = "BYPASS",
105         altsyncram_component.indata_reg_b = "CLOCK0",
106         altsyncram_component.intended_device_family = "Cyclone IV E",
107         altsyncram_component.lpm_type = "altsyncram",
108         altsyncram_component.numwords_a = 256,

```



```

109     altsyncram_component.numwords_b = 256,
110     altsyncram_component.operation_mode = "BIDIR_DUAL_PORT",
111     altsyncram_component.outdata_aclr_a = "NONE",
112     altsyncram_component.outdata_aclr_b = "NONE",
113     altsyncram_component.outdata_reg_a = "UNREGISTERED",
114     altsyncram_component.outdata_reg_b = "UNREGISTERED",
115     altsyncram_component.power_up_uninitialized = "FALSE",
116     altsyncram_component.ram_block_type = "M9K",
117     altsyncram_component.read_during_write_mode_mixed_ports = "DONT_CARE",
118     altsyncram_component.read_during_write_mode_port_a = "NEW_DATA_WITH_NBE_READ",
119     altsyncram_component.read_during_write_mode_port_b = "NEW_DATA_WITH_NBE_READ",
120     altsyncram_component.widthad_a = 8,
121     altsyncram_component.widthad_b = 8,
122     altsyncram_component.width_a = 6,
123     altsyncram_component.width_b = 6,
124     altsyncram_component.width_byteena_a = 1,
125     altsyncram_component.width_byteena_b = 1,
126     altsyncram_component.wrcontrol_wraddress_reg_b = "CLOCK0";
127
128
129 endmodule
130
131 // =====
132 // CNX file retrieval info
133 // =====
134 // Retrieval info: PRIVATE: ADDRESSSTALL_A NUMERIC "0"
135 // Retrieval info: PRIVATE: ADDRESSSTALL_B NUMERIC "0"
136 // Retrieval info: PRIVATE: BYTEENA_ACLR_A NUMERIC "0"
137 // Retrieval info: PRIVATE: BYTEENA_ACLR_B NUMERIC "0"
138 // Retrieval info: PRIVATE: BYTE_ENABLE_A NUMERIC "0"
139 // Retrieval info: PRIVATE: BYTE_ENABLE_B NUMERIC "0"
140 // Retrieval info: PRIVATE: BYTE_SIZE NUMERIC "8"
141 // Retrieval info: PRIVATE: BlankMemory NUMERIC "1"
142 // Retrieval info: PRIVATE: CLOCK_ENABLE_INPUT_A NUMERIC "0"
143 // Retrieval info: PRIVATE: CLOCK_ENABLE_INPUT_B NUMERIC "0"
144 // Retrieval info: PRIVATE: CLOCK_ENABLE_OUTPUT_A NUMERIC "0"
145 // Retrieval info: PRIVATE: CLOCK_ENABLE_OUTPUT_B NUMERIC "0"
146 // Retrieval info: PRIVATE: CLRdata NUMERIC "0"
147 // Retrieval info: PRIVATE: CLRq NUMERIC "0"
148 // Retrieval info: PRIVATE: CLRrdaddress NUMERIC "0"
149 // Retrieval info: PRIVATE: CLRrren NUMERIC "0"
150 // Retrieval info: PRIVATE: CLRwraddress NUMERIC "0"
151 // Retrieval info: PRIVATE: CLRwren NUMERIC "0"
152 // Retrieval info: PRIVATE: Clock NUMERIC "0"
153 // Retrieval info: PRIVATE: Clock_A NUMERIC "0"
154 // Retrieval info: PRIVATE: Clock_B NUMERIC "0"
155 // Retrieval info: PRIVATE: IMPLEMENT_IN_LES NUMERIC "0"
156 // Retrieval info: PRIVATE: INDATA_ACLR_B NUMERIC "0"
157 // Retrieval info: PRIVATE: INDATA_REG_B NUMERIC "1"
158 // Retrieval info: PRIVATE: INIT_FILE_LAYOUT STRING "PORT_A"
159 // Retrieval info: PRIVATE: INIT_TO_SIM_X NUMERIC "0"
160 // Retrieval info: PRIVATE: INTENDED_DEVICE_FAMILY STRING "Cyclone IV E"
161 // Retrieval info: PRIVATE: JTAG_ENABLED NUMERIC "0"
162 // Retrieval info: PRIVATE: JTAG_ID STRING "NONE"
163 // Retrieval info: PRIVATE: MAXIMUM_DEPTH NUMERIC "0"
164 // Retrieval info: PRIVATE: MEMSIZE NUMERIC "1536"
165 // Retrieval info: PRIVATE: MEM_IN_BITS NUMERIC "0"
166 // Retrieval info: PRIVATE: MIFfilename STRING ""
167 // Retrieval info: PRIVATE: OPERATION_MODE NUMERIC "3"
168 // Retrieval info: PRIVATE: OUTDATA_ACLR_B NUMERIC "0"
169 // Retrieval info: PRIVATE: OUTDATA_REG_B NUMERIC "0"
170 // Retrieval info: PRIVATE: RAM_BLOCK_TYPE NUMERIC "2"
171 // Retrieval info: PRIVATE: READ_DURING_WRITE_MODE_MIXED_PORTS NUMERIC "2"
172 // Retrieval info: PRIVATE: READ_DURING_WRITE_MODE_PORT_A NUMERIC "4"
173 // Retrieval info: PRIVATE: READ_DURING_WRITE_MODE_PORT_B NUMERIC "4"
174 // Retrieval info: PRIVATE: REGdata NUMERIC "1"
175 // Retrieval info: PRIVATE: REGq NUMERIC "0"
176 // Retrieval info: PRIVATE: REGrdaddress NUMERIC "0"
177 // Retrieval info: PRIVATE: REGrren NUMERIC "0"
178 // Retrieval info: PRIVATE: REGwraddress NUMERIC "1"
179 // Retrieval info: PRIVATE: REGwren NUMERIC "1"
180 // Retrieval info: PRIVATE: SYNTH_WRAPPER_GEN_POSTFIX STRING "0"
181 // Retrieval info: PRIVATE: USE_DIFF_CLKEN NUMERIC "0"
182 // Retrieval info: PRIVATE: UseDPRAM NUMERIC "1"
183 // Retrieval info: PRIVATE: VarWidth NUMERIC "0"
184 // Retrieval info: PRIVATE: WIDTH_READ_A NUMERIC "6"

```

```

185 // Retrieval info: PRIVATE: WIDTH_READ_B NUMERIC "6"
186 // Retrieval info: PRIVATE: WIDTH_WRITE_A NUMERIC "6"
187 // Retrieval info: PRIVATE: WIDTH_WRITE_B NUMERIC "6"
188 // Retrieval info: PRIVATE: WRADDR_ACLR_B NUMERIC "0"
189 // Retrieval info: PRIVATE: WRADDR_REG_B NUMERIC "1"
190 // Retrieval info: PRIVATE: WRCTRL_ACLR_B NUMERIC "0"
191 // Retrieval info: PRIVATE: enable NUMERIC "0"
192 // Retrieval info: PRIVATE: rden NUMERIC "0"
193 // Retrieval info: LIBRARY: altera_mf altera_mf.altera_mf_components.all
194 // Retrieval info: CONSTANT: ADDRESS_REG_B STRING "CLOCK0"
195 // Retrieval info: CONSTANT: CLOCK_ENABLE_INPUT_A STRING "BYPASS"
196 // Retrieval info: CONSTANT: CLOCK_ENABLE_INPUT_B STRING "BYPASS"
197 // Retrieval info: CONSTANT: CLOCK_ENABLE_OUTPUT_A STRING "BYPASS"
198 // Retrieval info: CONSTANT: CLOCK_ENABLE_OUTPUT_B STRING "BYPASS"
199 // Retrieval info: CONSTANT: INDATA_REG_B STRING "CLOCK0"
200 // Retrieval info: CONSTANT: INTENDED_DEVICE_FAMILY STRING "Cyclone IV E"
201 // Retrieval info: CONSTANT: LPM_TYPE STRING "altsyncram"
202 // Retrieval info: CONSTANT: NUMWORDS_A NUMERIC "256"
203 // Retrieval info: CONSTANT: NUMWORDS_B NUMERIC "256"
204 // Retrieval info: CONSTANT: OPERATION_MODE STRING "BIDIR_DUAL_PORT"
205 // Retrieval info: CONSTANT: OUTDATA_ACLR_A STRING "NONE"
206 // Retrieval info: CONSTANT: OUTDATA_ACLR_B STRING "NONE"
207 // Retrieval info: CONSTANT: OUTDATA_REG_A STRING "UNREGISTERED"
208 // Retrieval info: CONSTANT: OUTDATA_REG_B STRING "UNREGISTERED"
209 // Retrieval info: CONSTANT: POWER_UP_UNINITIALIZED STRING "FALSE"
210 // Retrieval info: CONSTANT: RAM_BLOCK_TYPE STRING "M9K"
211 // Retrieval info: CONSTANT: READ_DURING_WRITE_MODE_MIXED_PORTS STRING "DONT_CARE"
212 // Retrieval info: CONSTANT: READ_DURING_WRITE_MODE_PORT_A STRING "NEW_DATA_WITH_NBE_READ"
213 // Retrieval info: CONSTANT: READ_DURING_WRITE_MODE_PORT_B STRING "NEW_DATA_WITH_NBE_READ"
214 // Retrieval info: CONSTANT: WIDTHAD_A NUMERIC "8"
215 // Retrieval info: CONSTANT: WIDTHAD_B NUMERIC "8"
216 // Retrieval info: CONSTANT: WIDTH_A NUMERIC "6"
217 // Retrieval info: CONSTANT: WIDTH_B NUMERIC "6"
218 // Retrieval info: CONSTANT: WIDTH_BYTEENA_A NUMERIC "1"
219 // Retrieval info: CONSTANT: WIDTH_BYTEENA_B NUMERIC "1"
220 // Retrieval info: CONSTANT: WRCONTROL_WRAADDRESS_REG_B STRING "CLOCK0"
221 // Retrieval info: USED_PORT: address_a 0 0 8 0 INPUT NODEFVAL "address_a[7..0]"
222 // Retrieval info: USED_PORT: address_b 0 0 8 0 INPUT NODEFVAL "address_b[7..0]"
223 // Retrieval info: USED_PORT: clock 0 0 0 0 INPUT VCC "clock"
224 // Retrieval info: USED_PORT: data_a 0 0 6 0 INPUT NODEFVAL "data_a[5..0]"
225 // Retrieval info: USED_PORT: data_b 0 0 6 0 INPUT NODEFVAL "data_b[5..0]"
226 // Retrieval info: USED_PORT: q_a 0 0 6 0 OUTPUT NODEFVAL "q_a[5..0]"
227 // Retrieval info: USED_PORT: q_b 0 0 6 0 OUTPUT NODEFVAL "q_b[5..0]"
228 // Retrieval info: USED_PORT: wren_a 0 0 0 0 INPUT GND "wren_a"
229 // Retrieval info: USED_PORT: wren_b 0 0 0 0 INPUT GND "wren_b"
230 // Retrieval info: CONNECT: @address_a 0 0 8 0 address_a 0 0 8 0
231 // Retrieval info: CONNECT: @address_b 0 0 8 0 address_b 0 0 8 0
232 // Retrieval info: CONNECT: @clock0 0 0 0 0 clock 0 0 0 0
233 // Retrieval info: CONNECT: @data_a 0 0 6 0 data_a 0 0 6 0
234 // Retrieval info: CONNECT: @data_b 0 0 6 0 data_b 0 0 6 0
235 // Retrieval info: CONNECT: @wren_a 0 0 0 0 wren_a 0 0 0 0
236 // Retrieval info: CONNECT: @wren_b 0 0 0 0 wren_b 0 0 0 0
237 // Retrieval info: CONNECT: q_a 0 0 6 0 @q_a 0 0 6 0
238 // Retrieval info: CONNECT: q_b 0 0 6 0 @q_b 0 0 6 0
239 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port.v TRUE
240 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port.inc FALSE
241 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port.cmp FALSE
242 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port.bsf FALSE
243 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port_inst.v FALSE
244 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port_bb.v TRUE
245 // Retrieval info: LIB_FILE: altera_mf

```

A.5. Quartus RAM Black Box: ram_2port_bb.v

```

1 // megafunction wizard: %RAM: 2-PORT%VBB%
2 // GENERATION: STANDARD
3 // VERSION: WM1.0
4 // MODULE: altsyncram
5
6 // =====
7 // File Name: ram_2port.v
8 // Megafunction Name(s):
9 //   altsyncram
10 //

```

```

11 // Simulation Library Files(s):
12 //   altera_mf
13 // =====
14 // *****
15 // THIS IS A WIZARD-GENERATED FILE. DO NOT EDIT THIS FILE!
16 //
17 // 25.1std.0 Build 1129 10/21/2025 SC Lite Edition
18 // *****
19
20 //Copyright (C) 2025 Altera Corporation. All rights reserved.
21 //Your use of Altera Corporation's design tools, logic functions
22 //and other software and tools, and any partner logic
23 //functions, and any output files from any of the foregoing
24 //(including device programming or simulation files), and any
25 //associated documentation or information are expressly subject
26 //to the terms and conditions of the Altera Program License
27 //Subscription Agreement, the Altera Quartus Prime License Agreement,
28 //the Altera IP License Agreement, or other applicable license
29 //agreement, including, without limitation, that your use is for
30 //the sole purpose of programming logic devices manufactured by
31 //Altera and sold by Altera or its authorized distributors. Please
32 //refer to the Altera Software License Subscription Agreements
33 //on the Quartus Prime software download page.
34
35 module ram_2port (
36     address_a,
37     address_b,
38     clock,
39     data_a,
40     data_b,
41     wren_a,
42     wren_b,
43     q_a,
44     q_b);
45
46     input [7:0] address_a;
47     input [7:0] address_b;
48     input clock;
49     input [5:0] data_a;
50     input [5:0] data_b;
51     input wren_a;
52     input wren_b;
53     output [5:0] q_a;
54     output [5:0] q_b;
55     `ifndef ALTERA_RESERVED_QIS
56 // synopsys translate_off
57     `endif
58     tri1 clock;
59     tri0 wren_a;
60     tri0 wren_b;
61     `ifndef ALTERA_RESERVED_QIS
62 // synopsys translate_on
63     `endif
64
65 endmodule
66
67 // =====
68 // CNX file retrieval info
69 // =====
70 // Retrieval info: PRIVATE: ADDRESSSTALL_A NUMERIC "0"
71 // Retrieval info: PRIVATE: ADDRESSSTALL_B NUMERIC "0"
72 // Retrieval info: PRIVATE: BYTEENA_ACLR_A NUMERIC "0"
73 // Retrieval info: PRIVATE: BYTEENA_ACLR_B NUMERIC "0"
74 // Retrieval info: PRIVATE: BYTE_ENABLE_A NUMERIC "0"
75 // Retrieval info: PRIVATE: BYTE_ENABLE_B NUMERIC "0"
76 // Retrieval info: PRIVATE: BYTE_SIZE NUMERIC "8"
77 // Retrieval info: PRIVATE: BlankMemory NUMERIC "1"
78 // Retrieval info: PRIVATE: CLOCK_ENABLE_INPUT_A NUMERIC "0"
79 // Retrieval info: PRIVATE: CLOCK_ENABLE_INPUT_B NUMERIC "0"
80 // Retrieval info: PRIVATE: CLOCK_ENABLE_OUTPUT_A NUMERIC "0"
81 // Retrieval info: PRIVATE: CLOCK_ENABLE_OUTPUT_B NUMERIC "0"
82 // Retrieval info: PRIVATE: CLRdata NUMERIC "0"
83 // Retrieval info: PRIVATE: CLRq NUMERIC "0"
84 // Retrieval info: PRIVATE: CLRrdaddress NUMERIC "0"
85 // Retrieval info: PRIVATE: CLRrren NUMERIC "0"
86 // Retrieval info: PRIVATE: CLRwraddress NUMERIC "0"

```

```

87 // Retrieval info: PRIVATE: CLRwren NUMERIC "0"
88 // Retrieval info: PRIVATE: Clock NUMERIC "0"
89 // Retrieval info: PRIVATE: Clock_A NUMERIC "0"
90 // Retrieval info: PRIVATE: Clock_B NUMERIC "0"
91 // Retrieval info: PRIVATE: IMPLEMENT_IN_LES NUMERIC "0"
92 // Retrieval info: PRIVATE: INDATA_ACLR_B NUMERIC "0"
93 // Retrieval info: PRIVATE: INDATA_REG_B NUMERIC "1"
94 // Retrieval info: PRIVATE: INIT_FILE_LAYOUT STRING "PORT_A"
95 // Retrieval info: PRIVATE: INIT_TO_SIM_X NUMERIC "0"
96 // Retrieval info: PRIVATE: INTENDED_DEVICE_FAMILY STRING "Cyclone IV E"
97 // Retrieval info: PRIVATE: JTAG_ENABLED NUMERIC "0"
98 // Retrieval info: PRIVATE: JTAG_ID STRING "NONE"
99 // Retrieval info: PRIVATE: MAXIMUM_DEPTH NUMERIC "0"
100 // Retrieval info: PRIVATE: MEMSIZE NUMERIC "1536"
101 // Retrieval info: PRIVATE: MEM_IN_BITS NUMERIC "0"
102 // Retrieval info: PRIVATE: MIFfilename STRING ""
103 // Retrieval info: PRIVATE: OPERATION_MODE NUMERIC "3"
104 // Retrieval info: PRIVATE: OUTDATA_ACLR_B NUMERIC "0"
105 // Retrieval info: PRIVATE: OUTDATA_REG_B NUMERIC "0"
106 // Retrieval info: PRIVATE: RAM_BLOCK_TYPE NUMERIC "2"
107 // Retrieval info: PRIVATE: READ_DURING_WRITE_MODE_MIXED_PORTS NUMERIC "2"
108 // Retrieval info: PRIVATE: READ_DURING_WRITE_MODE_PORT_A NUMERIC "4"
109 // Retrieval info: PRIVATE: READ_DURING_WRITE_MODE_PORT_B NUMERIC "4"
110 // Retrieval info: PRIVATE: REGdata NUMERIC "1"
111 // Retrieval info: PRIVATE: REGq NUMERIC "0"
112 // Retrieval info: PRIVATE: REGrdaddress NUMERIC "0"
113 // Retrieval info: PRIVATE: REGrren NUMERIC "0"
114 // Retrieval info: PRIVATE: REGwaddress NUMERIC "1"
115 // Retrieval info: PRIVATE: REGwren NUMERIC "1"
116 // Retrieval info: PRIVATE: SYNTH_WRAPPER_GEN_POSTFIX STRING "0"
117 // Retrieval info: PRIVATE: USE_DIFF_CLKEN NUMERIC "0"
118 // Retrieval info: PRIVATE: UseDPRAM NUMERIC "1"
119 // Retrieval info: PRIVATE: VarWidth NUMERIC "0"
120 // Retrieval info: PRIVATE: WIDTH_READ_A NUMERIC "6"
121 // Retrieval info: PRIVATE: WIDTH_READ_B NUMERIC "6"
122 // Retrieval info: PRIVATE: WIDTH_WRITE_A NUMERIC "6"
123 // Retrieval info: PRIVATE: WIDTH_WRITE_B NUMERIC "6"
124 // Retrieval info: PRIVATE: WRADDR_ACLR_B NUMERIC "0"
125 // Retrieval info: PRIVATE: WRADDR_REG_B NUMERIC "1"
126 // Retrieval info: PRIVATE: WRCTRL_ACLR_B NUMERIC "0"
127 // Retrieval info: PRIVATE: enable NUMERIC "0"
128 // Retrieval info: PRIVATE: rden NUMERIC "0"
129 // Retrieval info: LIBRARY: altera_mf altera_mf.altera_mf_components.all
130 // Retrieval info: CONSTANT: ADDRESS_REG_B STRING "CLOCK0"
131 // Retrieval info: CONSTANT: CLOCK_ENABLE_INPUT_A STRING "BYPASS"
132 // Retrieval info: CONSTANT: CLOCK_ENABLE_INPUT_B STRING "BYPASS"
133 // Retrieval info: CONSTANT: CLOCK_ENABLE_OUTPUT_A STRING "BYPASS"
134 // Retrieval info: CONSTANT: CLOCK_ENABLE_OUTPUT_B STRING "BYPASS"
135 // Retrieval info: CONSTANT: INDATA_REG_B STRING "CLOCK0"
136 // Retrieval info: CONSTANT: INTENDED_DEVICE_FAMILY STRING "Cyclone IV E"
137 // Retrieval info: CONSTANT: LPM_TYPE STRING "altsyncram"
138 // Retrieval info: CONSTANT: NUMWORDS_A NUMERIC "256"
139 // Retrieval info: CONSTANT: NUMWORDS_B NUMERIC "256"
140 // Retrieval info: CONSTANT: OPERATION_MODE STRING "BIDIR_DUAL_PORT"
141 // Retrieval info: CONSTANT: OUTDATA_ACLR_A STRING "NONE"
142 // Retrieval info: CONSTANT: OUTDATA_ACLR_B STRING "NONE"
143 // Retrieval info: CONSTANT: OUTDATA_REG_A STRING "UNREGISTERED"
144 // Retrieval info: CONSTANT: OUTDATA_REG_B STRING "UNREGISTERED"
145 // Retrieval info: CONSTANT: POWER_UP_UNINITIALIZED STRING "FALSE"
146 // Retrieval info: CONSTANT: RAM_BLOCK_TYPE STRING "M9K"
147 // Retrieval info: CONSTANT: READ_DURING_WRITE_MODE_MIXED_PORTS STRING "DONT_CARE"
148 // Retrieval info: CONSTANT: READ_DURING_WRITE_MODE_PORT_A STRING "NEW_DATA_WITH_NBE_READ"
149 // Retrieval info: CONSTANT: READ_DURING_WRITE_MODE_PORT_B STRING "NEW_DATA_WITH_NBE_READ"
150 // Retrieval info: CONSTANT: WIDTHAD_A NUMERIC "8"
151 // Retrieval info: CONSTANT: WIDTHAD_B NUMERIC "8"
152 // Retrieval info: CONSTANT: WIDTH_A NUMERIC "6"
153 // Retrieval info: CONSTANT: WIDTH_B NUMERIC "6"
154 // Retrieval info: CONSTANT: WIDTH_BYTEENA_A NUMERIC "1"
155 // Retrieval info: CONSTANT: WIDTH_BYTEENA_B NUMERIC "1"
156 // Retrieval info: CONSTANT: WRCONTROL_WADDRESS_REG_B STRING "CLOCK0"
157 // Retrieval info: USED_PORT: address_a 0 0 8 0 INPUT NODEFVAL "address_a[7..0]"
158 // Retrieval info: USED_PORT: address_b 0 0 8 0 INPUT NODEFVAL "address_b[7..0]"
159 // Retrieval info: USED_PORT: clock 0 0 0 0 INPUT VCC "clock"
160 // Retrieval info: USED_PORT: data_a 0 0 6 0 INPUT NODEFVAL "data_a[5..0]"
161 // Retrieval info: USED_PORT: data_b 0 0 6 0 INPUT NODEFVAL "data_b[5..0]"
162 // Retrieval info: USED_PORT: q_a 0 0 6 0 OUTPUT NODEFVAL "q_a[5..0]"

```

```

163 // Retrieval info: USED_PORT: q_b 0 0 6 0 OUTPUT NODEFVAL "q_b[5..0]"
164 // Retrieval info: USED_PORT: wren_a 0 0 0 0 INPUT GND "wren_a"
165 // Retrieval info: USED_PORT: wren_b 0 0 0 0 INPUT GND "wren_b"
166 // Retrieval info: CONNECT: @address_a 0 0 8 0 address_a 0 0 8 0
167 // Retrieval info: CONNECT: @address_b 0 0 8 0 address_b 0 0 8 0
168 // Retrieval info: CONNECT: @clock0 0 0 0 0 clock 0 0 0 0
169 // Retrieval info: CONNECT: @data_a 0 0 6 0 data_a 0 0 6 0
170 // Retrieval info: CONNECT: @data_b 0 0 6 0 data_b 0 0 6 0
171 // Retrieval info: CONNECT: @wren_a 0 0 0 0 wren_a 0 0 0 0
172 // Retrieval info: CONNECT: @wren_b 0 0 0 0 wren_b 0 0 0 0
173 // Retrieval info: CONNECT: q_a 0 0 6 0 @q_a 0 0 6 0
174 // Retrieval info: CONNECT: q_b 0 0 6 0 @q_b 0 0 6 0
175 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port.v TRUE
176 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port.inc FALSE
177 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port.cmp FALSE
178 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port.bsf FALSE
179 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port_inst.v FALSE
180 // Retrieval info: GEN_FILE: TYPE_NORMAL ram_2port_bb.v TRUE
181 // Retrieval info: LIB_FILE: altera_mf

```

B. Pin Assignment Tcl File

```

1  # DE2-115 Cyclone IV E pin assignment
2
3  # HEX0
4  set_location_assignment PIN_H22 -to HEX0[6]
5  set_location_assignment PIN_J22 -to HEX0[5]
6  set_location_assignment PIN_L25 -to HEX0[4]
7  set_location_assignment PIN_L26 -to HEX0[3]
8  set_location_assignment PIN_E17 -to HEX0[2]
9  set_location_assignment PIN_F22 -to HEX0[1]
10 set_location_assignment PIN_G18 -to HEX0[0]
11
12 # HEX1
13 set_location_assignment PIN_U24 -to HEX1[6]
14 set_location_assignment PIN_U23 -to HEX1[5]
15 set_location_assignment PIN_W25 -to HEX1[4]
16 set_location_assignment PIN_W22 -to HEX1[3]
17 set_location_assignment PIN_W21 -to HEX1[2]
18 set_location_assignment PIN_Y22 -to HEX1[1]
19 set_location_assignment PIN_M24 -to HEX1[0]
20
21 # HEX2
22 set_location_assignment PIN_W28 -to HEX2[6]
23 set_location_assignment PIN_W27 -to HEX2[5]
24 set_location_assignment PIN_Y26 -to HEX2[4]
25 set_location_assignment PIN_W26 -to HEX2[3]
26 set_location_assignment PIN_Y25 -to HEX2[2]
27 set_location_assignment PIN_AA26 -to HEX2[1]
28 set_location_assignment PIN_AA25 -to HEX2[0]
29
30 # HEX3
31 set_location_assignment PIN_Y19 -to HEX3[6]
32 set_location_assignment PIN_AF23 -to HEX3[5]
33 set_location_assignment PIN_AD24 -to HEX3[4]
34 set_location_assignment PIN_AA21 -to HEX3[3]
35 set_location_assignment PIN_AB20 -to HEX3[2]
36 set_location_assignment PIN_U21 -to HEX3[1]
37 set_location_assignment PIN_V21 -to HEX3[0]
38
39 # HEX4
40 set_location_assignment PIN_AE18 -to HEX4[6]
41 set_location_assignment PIN_AF19 -to HEX4[5]
42 set_location_assignment PIN_AE19 -to HEX4[4]
43 set_location_assignment PIN_AH21 -to HEX4[3]
44 set_location_assignment PIN_AG21 -to HEX4[2]
45 set_location_assignment PIN_AA19 -to HEX4[1]
46 set_location_assignment PIN_AB19 -to HEX4[0]
47
48 # HEX5
49 set_location_assignment PIN_AH18 -to HEX5[6]
50 set_location_assignment PIN_AF18 -to HEX5[5]
51 set_location_assignment PIN_AG19 -to HEX5[4]

```

```

52 set_location_assignment PIN_AH19 -to HEX5[3]
53 set_location_assignment PIN_AB18 -to HEX5[2]
54 set_location_assignment PIN_AC18 -to HEX5[1]
55 set_location_assignment PIN_AD18 -to HEX5[0]
56
57 # HEX6
58 set_location_assignment PIN_AC17 -to HEX6[6]
59 set_location_assignment PIN_AA15 -to HEX6[5]
60 set_location_assignment PIN_AB15 -to HEX6[4]
61 set_location_assignment PIN_AB17 -to HEX6[3]
62 set_location_assignment PIN_AA16 -to HEX6[2]
63 set_location_assignment PIN_AB16 -to HEX6[1]
64 set_location_assignment PIN_AA17 -to HEX6[0]
65
66 # HEX7
67 set_location_assignment PIN_AA14 -to HEX7[6]
68 set_location_assignment PIN_AG18 -to HEX7[5]
69 set_location_assignment PIN_AF17 -to HEX7[4]
70 set_location_assignment PIN_AH17 -to HEX7[3]
71 set_location_assignment PIN_AG17 -to HEX7[2]
72 set_location_assignment PIN_AE17 -to HEX7[1]
73 set_location_assignment PIN_AD17 -to HEX7[0]
74
75 # LEDGO = write mode indicator
76 set_location_assignment PIN_E21 -to LEDGO
77
78 # Switches
79 set_location_assignment PIN_AB28 -to SW[0]
80 set_location_assignment PIN_AC28 -to SW[1]
81 set_location_assignment PIN_AC27 -to SW[2]
82 set_location_assignment PIN_AD27 -to SW[3]
83 set_location_assignment PIN_AB27 -to SW[4]
84 set_location_assignment PIN_AC26 -to SW[5]
85 set_location_assignment PIN_AD26 -to SW[6]
86 set_location_assignment PIN_AB26 -to SW[7]
87 set_location_assignment PIN_AC25 -to SW[8]
88 set_location_assignment PIN_AB25 -to SW[9]
89 set_location_assignment PIN_AC24 -to SW[10]
90 set_location_assignment PIN_AB24 -to SW[11]
91 set_location_assignment PIN_AB23 -to SW[12]
92 set_location_assignment PIN_AA24 -to SW[13]
93 set_location_assignment PIN_AA23 -to SW[14]
94 set_location_assignment PIN_AA22 -to SW[15]
95 set_location_assignment PIN_Y24 -to SW[16]
96 set_location_assignment PIN_Y23 -to SW[17]
97
98 # 50 MHz clock
99 set_location_assignment PIN_Y2 -to CLOCK_50
100
101 # Keys
102 set_location_assignment PIN_M23 -to KEY[0]
103 set_location_assignment PIN_M21 -to KEY[1]
104 set_location_assignment PIN_N21 -to KEY[2]
105 set_location_assignment PIN_R24 -to KEY[3]

```

C. Conclusion

A dual-port memory of size $256 \text{ words} \times 6 \text{ bits}$ was configured using the Quartus Prime RAM: 2-PORT IP core and implemented on the Altera/Terasic DE2-115 FPGA. The generated `ram_2port.v` core was integrated with Verilog modules for top-level control, button pulse generation and seven-segment decoding. The final hardware demonstration verifies the intended write and read sequence for ports A and B and provides visible confirmation on the eight seven-segment displays.